Consider another function $f(x) = x^2 + 4, x \geq 0$. The function will be given by its graph. (See Figure 1.14)

Let x be a point in the domain. Let us draw a vertical line through the point x. Let it meet the curve at P. The point at which the horizontal line drawn through P meets the y-axis is $f(x)$. Similarly using horizontal lines through a point y in the co-domain, we can find the pre-images of y.

Can we say that any curve drawn on the plane be considered as a function from a subset of $\mathbb{R}$ to $\mathbb{R}$? No, we cannot. There is a simple test to find this.

Vertical Line Test

As we noted earlier, the vertical line through any point x in the domain meets the curve at some point, then the y-coordinate of the point is $f(x)$. If the vertical line through a point x in the domain meets the curve at more than one point, we will get more than one value for $f(x)$ for one x. This is not allowed in a function. Further, if the vertical line through a point x in the domain does not meet the curve, then there will be no image for x; this is also not possible in a function. So we can say,

> "if the vertical line through a point x in the domain meets the curve at more than one point or does not meet the curve, then the curve will not represent a function".

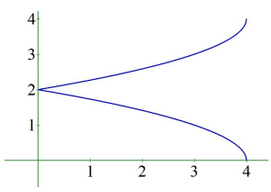


Figure 1.15

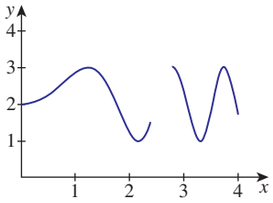


Figure 1.16

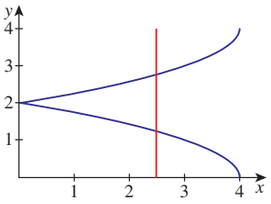


Figure 1.17

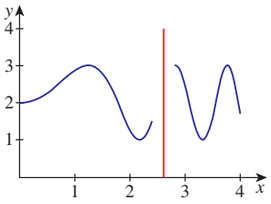


Figure 1.18

The curve indicated in Figure 1.15 does not represent a function from $[0, 4]$ to $\mathbb{R}$ because a vertical line meets the curve at more than one point (See Figure 1.17). The curve indicated in Figure 1.16 does not represent a function from $[0, 4]$ to $\mathbb{R}$ because a vertical line drawn through $x = 2.5$ in $[0, 4]$ does not meet the curve (See Figure 1.18).

Testing whether a given curve represents a function or not by drawing vertical lines is called ***vertical line test*** or simply vertical test.

The third curve $y^2 = x$ in Illustration 1.2 fails in the vertical line test and hence it is not a function from $\mathbb{R}$ to $\mathbb{R}$.

1.6.2 Some Elementary Functions

Some frequently used functions are known by names. Let us list some of them.

(i) Let X be any non-empty set. The function $f : X \to X$ defined by $f(x) = x$ for all $x \in X$ is called the ***identity function*** on X (See Figure 1.19). It is denoted by I_X or I.

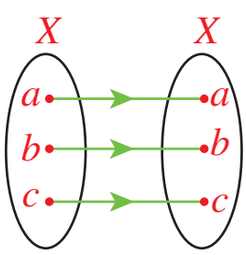


Figure 1.19

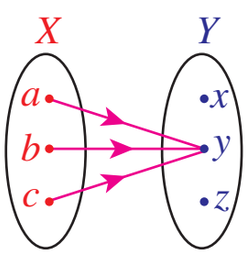


Figure 1.20